

Problems Proposed for 2025-26 Pal4Pal

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Eric

Q1 Geometry [4 points]

In $\triangle ABC$, the A -median, B -median, C -median have lengths 13, 14, 15, respectively. Find area of $\triangle ABC$.

Q2 Number Theory [4 points]

Given a positive base-10 positive integer $N = a_n a_{n-1} \dots a_1 a_0$, let $f(N) = 2^n a_0 + 2^{n-1} a_1 + \dots + 2^1 a_{n-1} + 2^0 a_n$. Find $f(f(\dots f(2014^{2014}) \dots))$ (2014^{2014} f 's).

Yichen

Q1 Geometry [4 points]

Let ABC be an equilateral triangle. Let P be a point inside, such that $PA = 6$, $PB = 12$, and $PC = 6\sqrt{3}$. Find the side length of the triangle.

Q2 Combo [4 points]

Kait rolls a fair 6-sided die until she rolls a 6. If she rolls a 6 on the N th roll, she then rolls the die N more times. What is the probability that she rolls a 6 during these next N times?

Justin

Q1 Geometry [4 points]

In triangle $\triangle ABC$ with orthocenter H , the internal angle bisector of $\angle BAC$ intersects $\overline{BC}$ at Y . Given that $AH = 4$, $AY = 6$, and the distance from Y to $\overline{AC}$ is $\sqrt{15}$, compute BC .

Q2 Number Theory [4 points]

Given that $20^{22} + 1$ has exactly 4 prime divisors $p_1 < p_2 < p_3 < p_4$, determine $p_1 + p_2$.